Michael Murray

Senior Mechanical Design Engineer | Product Development | DFM, FEA, Manufacturing

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Portfolio & More: mjm-murray.github.io | linkedin.com/in/michaeljohnmurray

SUMMARY

Mechanical engineer with 17 years of experience taking complex hardware from concept through production across medical devices, solar tracking systems, industrial power, and automation. Holds three U.S. patents in electromechanical design, with hands-on expertise in mechanism and device design, DFM, FEA, GD&T, and rapid prototyping. Leads cross-functional development and manufacturing transfer for safety-critical products.

PROFESSIONAL EXPERIENCE

Nevados Engineering | Oakland, CA

Senior Mechanical Design Engineer | April 2025 – October 2026

- Redesigned the stamped and welded tracker coupler assembly and took it through production, improving structural performance and manufacturability; sized welds per AWS D1.1 and AISC 360.
- Performed linear and nonlinear FEA (SOLIDWORKS Simulation) on weldments, couplers, torque tubes, and bolted joints under wind downforce, uplift, and torsion to validate factors of safety.
- Led DFMEA of the coupler assembly and validated designs, considering stiffness, ultimate strength, and fatigue. Conducted and analyzed lifecycle testing of production parts to verify design and quality.
- Drove cross-functional alignment through regular status reviews with the C-suite and functional leads.
- Owned fastener engineering and supplier quality: torque–tension and K-factor analysis (VDI 2230, ISO 16047), PPAP review, and supplier corrective actions; led fastener coating analysis and testing.
- Qualified gearmotor suppliers for slew drives, including torque-curve evaluation, splined-shaft interface review, and an output-shaft failure investigation.
- Created coating specs with suppliers for long-term outdoor exposure by site corrosivity; qualified via cyclic (IEC 60068-2-52) and continuous (ASTM B117) salt-spray testing, graded per ASTM D610.
- Standardized SOLIDWORKS templates and PDM workflows across the team.

Swope Design Solutions | San Francisco, CA

Senior Mechanical Engineer | June 2022 – April 2025

- Led design and build of a diaphragm pump-to-catheter system for removing calcified arterial lesions, from proof of concept through tested prototypes; served as customer-facing project manager.
- Designed and prototyped a precision positioning mechanism for a kidney-stone treatment device.
- Designed and built an automated data matrix scanning system, coding vision and analysis logic in C and Python, with a motor-driven positioning system for automated movement.
- Designed a heater subsystem for a medical device, performing thermal analysis and benchtop testing.
- Designed injection-molded parts and tooling, machined molds, and molded silicone parts.
- Redesigned endoscopic robot parts for injection molding, partnering with an overseas CM on moldability.
- Mentored junior engineers on design, DFM, and testing practices while keeping client programs on schedule.

True Digital Surgery | Santa Barbara, CA

Senior Staff Mechanical Design Engineer | November 2020 – December 2021

- Designed mechanical subsystems of a positioning system for a robotic neurosurgical 3D microscope.
- Performed FEA to prevent mechanical failures and confirm factors of safety per IEC 60601 requirements.
- Performed thermal analysis of heat-generating components to keep structures and electronics within allowable limits. Confirmed the heat-affected zone of weldments.
- Applied GD&T (ASME Y14.5) and tolerance stack-ups to ensure fit and function across assemblies.

Surround Medical Systems | Morrisville, NC

Senior Mechanical Engineer | May 2019 – September 2020

- Led the mechanical department in designing a 3D X-ray system from concept through production.
- Improved X-ray delivery to reduce patient radiation exposure while maintaining image quality.
- Co-authored a 510(k) submission for FDA clearance. Documented mechanical safety factor calcs and risks.
- Ran FEA and FMEA on armature and tube-positioning subsystems to identify and design out failure modes before production. Optimized collimator design for image quality.

ABB | Raleigh, NC
Senior Mechanical Engineer, R&D | March 2017 – May 2019

- Standardized transformer designs using common parts, reducing cost and lead times.
- Redesigned transformer parts and enclosure geometry to improve convective cooling and thermal margin.
- Led design automation of models, drawings, and BOM data; authored the team's programming standard.
- Developed an outdoor-rated transformer cabinet, specifying sealing for long-term environmental exposure.

Ingersoll Rand | Davidson, NC
Design Engineer III | August 2014 – March 2017

- Designed parts and assemblies for complex, custom industrial centrifugal air compressors (2,000–7,000 CFM, 400–1,250 HP) to meet ANSI and ISO standards.
- Designed a dual oil cooler system for 5,000 CFM air compressors.
- Specified components and geometry to achieve required pressure and flow rates with minimal losses through pre-build airflow assessments.

Newell Rubbermaid – Piedmont Hardware Brands | Charlotte, NC
Mechanical Design Engineer | February 2013 – August 2014

- Designed high-volume sheet metal, die-cast, injection-molded, and stamped parts.
- Used hand calcs and SOLIDWORKS Simulation FEA to identify failure modes early and guide design.
- Reduced unit cost of a complex door lock by 16% through DFM redesign in partnership with manufacturing.

Sandvik | Mebane, NC
Manufacturing Engineer | September 2011 – February 2013

- Reduced tooling costs by 30% via optimized CAM programs for 5-axis mills, lathes, and grinders.
- Created CAD models, CNC programs through CAM, and drawings for high-volume metal-cutting tools.
- Designed and implemented a coolant reclamation system to optimize machining efficiency and cut waste.

ABB | Lake Mary, FL
Mechanical Design Engineer | August 2009 – September 2011

- Invented a remotely operated device for industrial medium-voltage circuit breakers (5–15 kV); recognized with two patents (U.S. Patents 9,876,335 and 8,654,513).
- Developed a Modular Mechanism Operated Cell to retrofit circuit breakers (U.S. Patent 8,878,088).
- Redesigned circuit breaker airflow paths to reduce hot spots, eliminating overheating and braze-joint failures.

TECHNICAL SKILLS

DFM, DFA, DFx: Machined, stamped, welded, cast, sheet metal, and molded parts

Mechanical Design, CAD & Drafting: SOLIDWORKS, Creo, NX, Inventor, Onshape, Fusion

Finite Element Analysis (FEA): SOLIDWORKS Simulation (static, linear, nonlinear, modal, thermal)

Geometric Dimensioning & Tolerancing (GD&T): ASME Y14.5, tolerance stack analysis

Product Development & Project Management: Jira, Confluence, Monday.com, and more

Rapid Prototyping & 3D Printing: FDM, SLA, SLS, DMLS; CNC & manual machining; molding

Supplier Quality, Management, and Sourcing: PPAP, FAI, SCAR, qualification, supplier audits

Failure Mode & Effects Analysis (FMEA / DFMEA)

PDM / PLM: SOLIDWORKS PDM Professional administration, Arena, Windchill

Regulations & Standards: FDA, IEC 60601, ISO 13485, ASME, ASTM, ANSI, AWS, AISC, UL, IEEE

Thermal Analysis: SOLIDWORKS Simulation, Flow Simulation

CAM (Computer-Aided Manufacturing): HSMWorks, NX

Injection Molding & Plastics

EDUCATION

North Carolina State University, B.S. Mechanical Engineering, 2009
Honors: cum laude, Dean's List, 2008 NCSU ME Design Award, Co-Op Program

ADDITIONAL NOTES

Portfolio, Manufacturing Process Experience, and more: mjm-murray.github.io